

Department of Mechanical Engineering
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Fayetteville, AR, USA

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Editor-in-Chief
Applied Thermal Engineering

Dear Editor,

Please consider our manuscript, “BubbleID-Flow: Reproducible Machine-Vision Quantification of Projected Vapor Coverage in Subcooled Flow Boiling,” for publication as an Original Research Article in *Applied Thermal Engineering*.

The manuscript develops an auditable workflow for converting high-speed FC-72 flow-boiling images into two-dimensional projected vapor coverage. A one-class Mask R-CNN model fine-tuned on 130 annotated images is compared with fixed and learned pixel baselines. The analysis exposes the same-sequence holdout limitation, replaces independent-frame uncertainty with a moving-block bootstrap, reconstructs electrical heat flux and mass flux from documented geometry, and tests whether projected coverage improves held-out prediction of the measured surface-temperature response. It does not in this dataset, which defines a conservative and reproducible claim boundary. The work is aligned with *Applied Thermal Engineering* because it connects image-based measurement quality to thermal forcing, measured test-section temperatures, and validation requirements for flow-boiling diagnostics.

The corresponding author will confirm publication status, concurrent-submission status, author approval, funding, competing interests, and data-sharing language before this letter is used for submission.

Sincerely,

Han Hu
On behalf of the authors